Run report: weak-curve laboratory, run 20261013T0055Z

Descriptive record of one run. Interpretation is in the study `README.md` / `REPORT.pdf`.

What ran

- Command: `rustc -O weak_curve_lab.rs && ./weak_curve_lab all && ./weak_curve_lab e4x` (experiments E5, E1, E2, E3, E3b, E4, then E4x).
- Spec: `PROTOCOL.md`, frozen before the run. E4x (sextic with `a, b >= 1`) was added after E4 was first seen and is supplementary.
- Environment, base commit, source SHA-256, rustc version, host, load and wall time: `results/run-20261013T0055Z.env.txt` (reproduced below).
- Isolation: none. Shared host under heavy load; no timing is reported as a result. All recorded values are exact integer counts and are deterministic (fixed seed 20261012).
- Raw records: `results/run-20261013T0055Z.jsonl`, one JSON object per line, field `exp` names the experiment.

```
run: 20261013T0055Z
command: rustc -O weak_curve_lab.rs && ./weak_curve_lab all && ./weak_curve_lab e4x
commit (base): b5cc6ffcf00ee4da0f37ea6ba5735829d4e70c31
source sha256: fbd024b0e6ccaabc1176086a566b52d000ed9575dd8c102cfa20ddeb2ecf38d9
rustc: rustc 1.93.1 (01f6ddf75 2026-02-11) (Homebrew)
host: Darwin 25.6.0 arm64, Apple M4 Pro, 14 cpus
uptime at start: 17:55  up 1 day, 17:34, 10 users, load averages: 127.19 110.14 92.70
real 526.11
user 305.99
sys 15.98
uptime at end: 18:04  up 1 day, 17:43, 10 users, load averages: 109.71 127.08 117.01
```

Correction (added after review; PROTOCOL.md amendment 1)

The JSONL of this run holds **six** E4x records: `all` already ran E4x, and the command ran it again, so lines 120-122 repeat lines 117-119. Three distinct E4x runs completed; the table below counts those three and the JSONL should be read with lines 120-122 dropped. This is an accounting correction; no value changed. The corrected run is `20261013T0404Z`.

Runs requested and completed

Experiment	Requested	Completed	Notes
E5 paper data	2 examples	2	
E1 collapse + control	5 fields x (3 weak + 3 random)	30	no subgroup hit the cap
E2 lines	4 fields x 13 lines	52	
E3 census	6 fields x 4 curves	23	`x^5+c` not run at (5,4): singular in characteristic 5
E3b multi-form	6 curve/field pairs	6	union capped at 1.2e7 elements for (11,5) and (13,4); reported as lower bounds
E4 Kani	3 fields	3	
E4x Kani, supplementary	3 fields	3	not in the frozen protocol

Recorded values

All values are **measured** exact counts unless marked derived. Unit: number of group elements (`H`, `J`), number of pairs (`candidates`), number of lines.

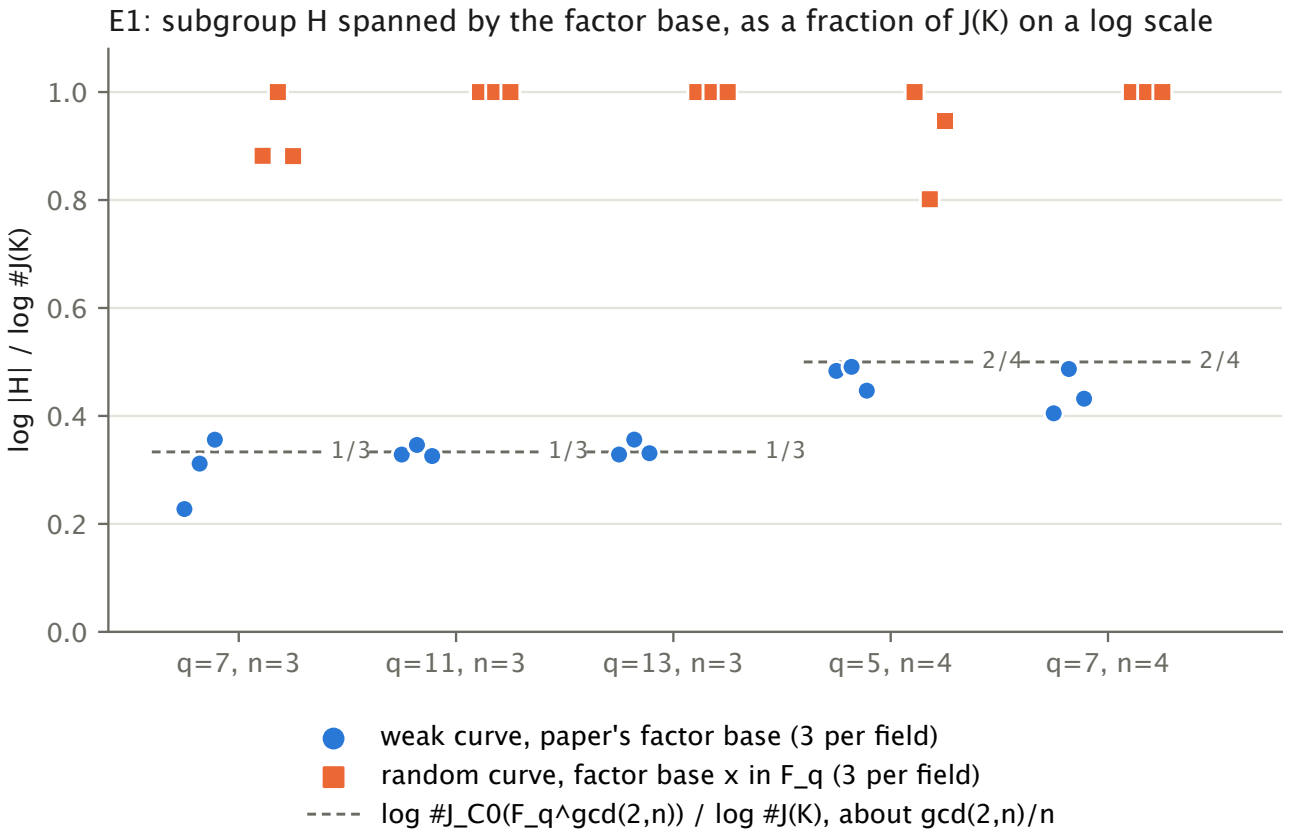


Figure 1: E1, run 20261013T0055Z. Each marker is one curve (n = 3 per field and kind). Vertical axis: log |H| / log #J(K), derived as a ratio of two measured counts. Dashed: the value gcd(2,n)/n. No uncertainty: counts are exact.

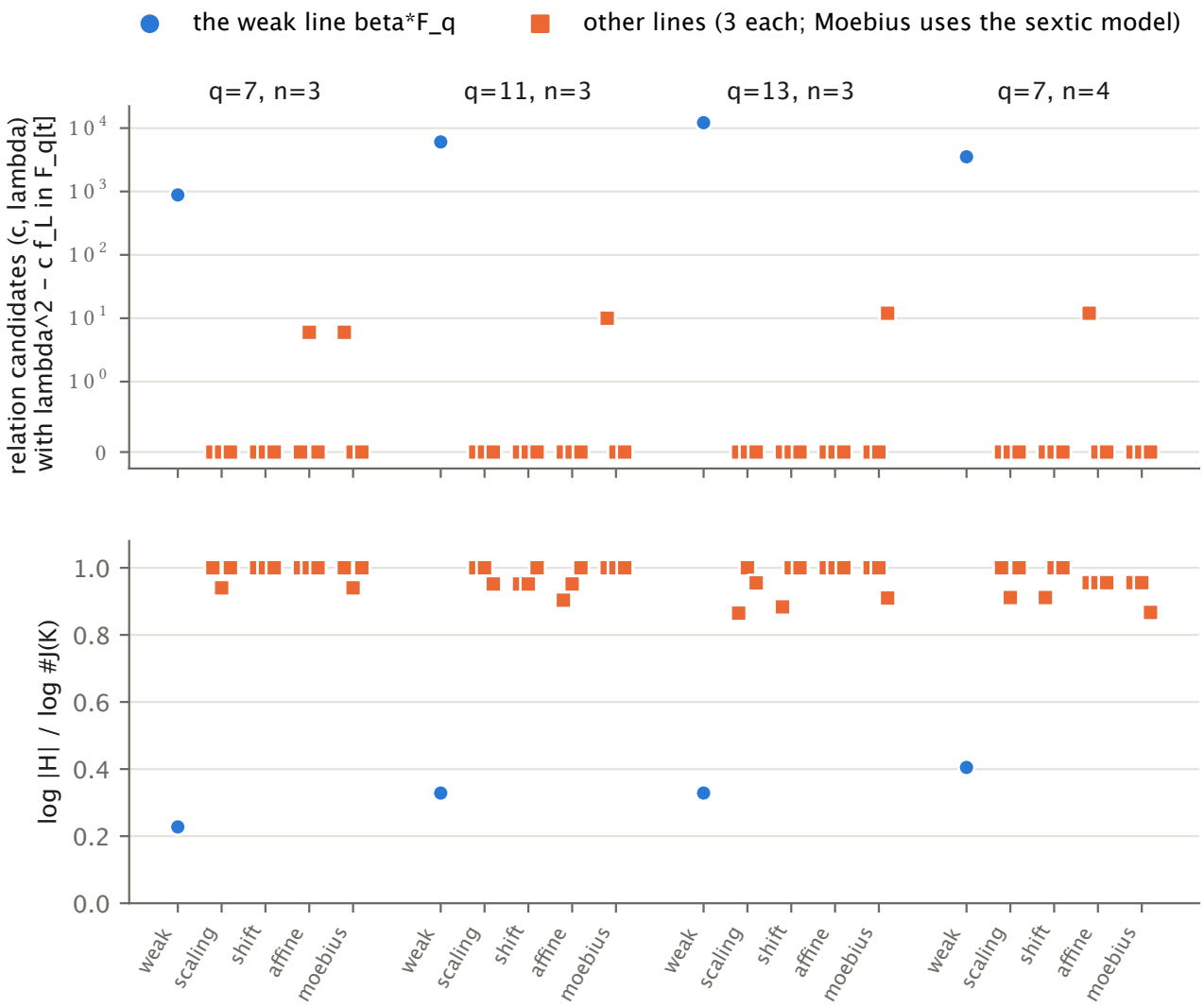


Figure 2: E2, run 20261013T0055Z. One weak curve per field, 13 factor-base lines each. Top: measured number of relation candidates (symlog axis; Moebius lines use the sextic model and are not like-for-like with the others). Bottom: log |H| / log #J(K).

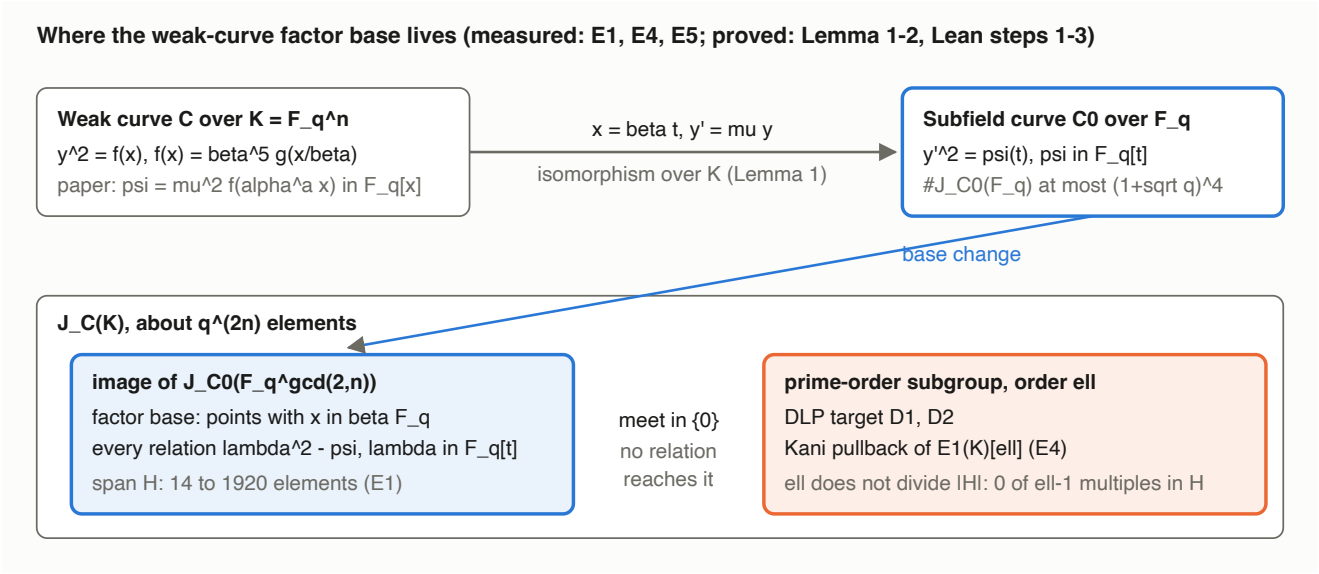


Figure 3: Diagram of what was run and where the measured subgroups sit.

Controls and cross-checks recorded in the run

- `orders_agree`: #J(K) from point counts over K and K^2 against det(1 - pi^n) from the F_q L-polynomial, on every weak curve and every Kani curve.

- `selftest`: #J(K) D = 0 and associativity on 20 random triples per E1 curve.
- `H_divides_sub`, `coords_in_subfield` (weak curves); `H_divides_J` (random curves).
- `hom_ok` / `hom_tested` (E4): homomorphism property of the pullback on random pairs.

The per-record values are in the JSONL; the study report tabulates them.

Missing data

- Union subgroup order for `x^5+c` at (11,5) and `x^5+cx` at (13,4): capped, lower bound recorded.
- Span of the second weak line found by the E3 census at (7,4): not computed.